

# Pre-Algebra





## About the Course

Students discover algebraic concepts using manipulatives.

This topic is included in the following course(s): Mathematics: Grade 7

### Mathematics: Grade 7

For students in elementary school, we recommend RightStart Math (Second Edition), because a) it adheres to Mason's principle that the idea always comes before the algorithm, b) it reflects the very best current research in how children learn mathematics, and c) it offers much-needed support for teachers whose own math instruction may have been lacking. Students in middle school take RightStart Geometry and also add in Pre-Algebra (Hands-On Equations) and Jacobs' Elementary Algebra.

The lessons should be done in the morning and games in the afternoon. Both mental math and games are critical parts of math instruction, so do not skip them. These lessons are paced out to help keep you on track. There are a few catch-up days built in and can be used to finish incomplete work or to review concepts. The general goal should be to complete one lesson per day. It is important that you complete one level per year, even if that means continuing during breaks, as moving slower can affect high school credits later.



## Placement & Combining Tips

Prerequisite: RightStart F or mastery of basic arithmetic, fractions, decimals, and percents.



## Scheduling

| GRADE | SCHEDULE INFO.                     | BOOKS                                                                             |
|-------|------------------------------------|-----------------------------------------------------------------------------------|
| 7     | Pre-Algebra<br>2 times/week 30 min | Hands-On Equations Deluxe Home Set<br>The Hands-On Equations Verbal Problems Book |

### Sample Weekly View

| Day 1                       | Day 2                                    | Day 3                                          | Day 4                                    | Day 5             |
|-----------------------------|------------------------------------------|------------------------------------------------|------------------------------------------|-------------------|
| <b>Mathematics: Grade 7</b> |                                          |                                                |                                          |                   |
| Geometry: Grade 7           | Pre-Algebra<br>Math Games: Grades<br>1-8 | Geometry: Grade 7<br>Math Games: Grades<br>1-8 | Pre-Algebra<br>Math Games: Grades<br>1-8 | Geometry: Grade 7 |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the

time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Mathematics: Grade 7

☐ If you downloaded the RightStart ebook, print out enough copies of each page and put them in a binder for easy access.

☐ Organize math manipulatives for easy access.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Pre-Algebra



Hands-On Equations Deluxe Home Set



The Hands-On Equations Verbal Problems Book



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

(No Quick Links Assigned)

[Click THIS text  
or scan the QR  
code for links.](#)



# Pre-Algebra

## How To Teach



### Read

The RightStart Math Level G materials are written to the student and may be done independently. However, you may still find it very helpful to work through the lessons to familiarize yourself with the concepts.



### Prepare

For Pre-Algebra, it is highly recommended that students be guided by a teacher during these lessons. Teachers may watch the video lessons to learn the concepts and how to teach them to students. If you choose to have your student work independently using the video lessons, you may still find it very helpful to work through the lessons yourself to familiarize yourself with the concepts. Teachers must stay apprised of how students are progressing and be ready to step in and help, since problems can compound quickly. If you are working through the lessons ahead of your student, note the places where you have to stop and think, because those are probably going to be places where students are going to be challenged, as well.



### Growth Mindset

Resist the temptation to fall back on just teaching a memorizable procedure when the student struggles with a concept. Growth happens in the struggle. Work on resilience by modeling a growth mindset.



### Leave & Return

When your student hits a wall during a lesson, stop and move on to the next subject in your day or take a break. Explain that even when you stop working on a problem, your brain continues to work on it while you do other things, and you don't even know it. Many times, when you come back to the problem the next day, it comes more easily.



### Connect

Look for ways to bring math into everyday life and conversation "by the way" - through cooking, gardening, house projects, health, games, etc. Students should discover how math is useful and beautiful, and how it helps us explain and navigate the world.



## Term 1

### WEEK 1 □ 30m Pre-Algebra - Lesson 1

*Balancing the Scale*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.  
Introduction & Level I  
Lesson 1 p.3-4

#### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 1

### WEEK 1 □ 30m Pre-Algebra - Lesson 2

*Balancing the Scale (continued)*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.  
Introduction & Level I  
Lesson 2 p.4

#### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 2

### WEEK 2 □ 30m Pre-Algebra - Lesson 3

*"Legal Moves"*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.  
Introduction & Level I  
Lesson 3 p.4-6

#### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 3

### WEEK 2 □ 30m Pre-Algebra - Lesson 4

*Subtracting from Both Sides*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.  
Introduction & Level I  
Lesson 4 p.6-7

#### → PRACTICE



## Term 1

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 4

### WEEK 3 30m Pre-Algebra - Lesson 5

*Original Physical Setup*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.  
Introduction & Level I  
Lesson 5 p.7-8

#### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 5

### WEEK 3 30m Pre-Algebra - Lesson 6

*Parentheses*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.  
Introduction & Level I  
Lesson 6 p.8

#### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 6

### WEEK 4 30m Pre-Algebra - Lesson 7

*Pictorial Notation*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.  
Introduction & Level I  
Lesson 7 p.9-10

#### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 7

### WEEK 4 30m Pre-Algebra - Lesson 8

*Getting Started with Verbal Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.  
Sec.1 p.8  
Problems 1-10



## Term 1

### WEEK 5 30m Pre-Algebra - Lesson 9

*Getting Started with Verbal Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.1 p.9

Problems 11-20

### WEEK 5 30m Pre-Algebra - Lesson 10

*Getting Started with Verbal Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.1 p.10

Problems 21-30

### WEEK 6 30m Pre-Algebra - Lesson 11

*Getting Started with Verbal Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.1 p.11

Problems 31-39

### WEEK 6 30m Pre-Algebra - Lesson 12

*Number Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.2 p.25

Problems 1-6

### WEEK 7 30m Pre-Algebra - Lesson 13

*Number Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.2 p.26

Problems 7-14

### WEEK 7 30m Pre-Algebra - Lesson 14

*Consecutive Number Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.



## Term 1

Sec.3 p.42  
Problems 1-11

### WEEK 8 30m Pre-Algebra - Lesson 15

*Age Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.4 p.57

Problems 1-10

### WEEK 8 30m Pre-Algebra - Lesson 16

*Age Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.4 p.57-58

Problems 11-19

### WEEK 9 30m Pre-Algebra - Lesson 17

*Coin Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.5 p.75-76

Problems 1-11

### WEEK 9 30m Pre-Algebra - Lesson 18

*Coin Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.5 p.77-78

Problems 12-26

### WEEK 10 30m Pre-Algebra - Lesson 19

*Distance Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.6 p.100-101

Problems 1-13





## Term 1

### WEEK 10 30m Pre-Algebra - Lesson 20

*Distance Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.6 p.101-103

Problems 14-26

### WEEK 11 30m Pre-Algebra - Lesson 21

*Story and Misc. Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.7 p.135

Problems 1-8

### WEEK 11 30m Pre-Algebra - Lesson 22

*Story and Misc. Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.7 p.136

Problems 9-15

### WEEK 12 30m Pre-Algebra - Lesson 23

*Exam Week*

Materials: Hands-On Equations

#### → EXAMS

Answer question(s) related to course.

Questions will come from:

Hands-On Equations

### WEEK 12 30m Pre-Algebra - Lesson 24

*Exam Week*

Materials: Hands-On Equations

#### → EXAMS

Answer question(s) related to course.

Questions will come from:

Hands-On Equations



## Term 2

### WEEK 1 30m Pre-Algebra - Lesson 25

*Introducing the Star*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 8 p.1-2

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 8

### WEEK 1 30m Pre-Algebra - Lesson 26

*Negatives*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 9 p.2-3

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 9

### WEEK 2 30m Pre-Algebra - Lesson 27

*Removing a Pair of Opposites*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 10 p.4

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 10

### WEEK 2 30m Pre-Algebra - Lesson 28

*Balancing by Adding*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 11 p.5

#### → PRACTICE



## Term 2

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 11

### WEEK 3 30m Pre-Algebra - Lesson 29

*Adding Pairs of Opposites*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 12 p.6

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 12

### WEEK 3 30m Pre-Algebra - Lesson 30

*Removing Quantities*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 13 p.7

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 13

### WEEK 4 30m Pre-Algebra - Lesson 31

*Adding Zero*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 14 p.8-9

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 14

### WEEK 4 30m Pre-Algebra - Lesson 32

*-x*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level II

Lesson 15 p.10



## Term 2

### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 15

### WEEK 5 30m Pre-Algebra - Lesson 33

*Pictorial Notation*

Materials: Hands-On Equations

### → DEMONSTRATE

Work through examples and discuss.  
Level II  
Lesson 16 p.11-13

### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 16

### WEEK 5 30m Pre-Algebra - Lesson 34

*Number Problems*

Materials: Hands-On Equations Verbal Problems

### → PRACTICE

Solve problems and check solutions.  
Sec.2 p.27  
Problems 15-19

### WEEK 6 30m Pre-Algebra - Lesson 35

*Number Problems (continued)*

Materials: Hands-On Equations Verbal Problems

### → PRACTICE

Solve problems and check solutions.  
Sec.2 p.27  
Problems 20-24

### WEEK 6 30m Pre-Algebra - Lesson 36

*Consecutive Number Problems*

Materials: Hands-On Equations Verbal Problems

### → PRACTICE

Solve problems and check solutions.  
Sec.3 p.43  
Problems 12-15

### WEEK 7 30m Pre-Algebra - Lesson 37

*Consecutive Number Problems (continued)*

Materials: Hands-On Equations Verbal Problems

### → PRACTICE

Solve problems and check solutions.  
Sec.3 p.43



## Term 2

Problems 16-19

### WEEK 7 30m Pre-Algebra - Lesson 38

*Age Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.4 p.59

Problems 20-25

### WEEK 8 30m Pre-Algebra - Lesson 39

*Coin Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.5 p.79

Problems 27-31

### WEEK 8 30m Pre-Algebra - Lesson 40

*Distance Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.6 p.104

Problems 27-30

### WEEK 9 30m Pre-Algebra - Lesson 41

*Distance Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.6 p.104-105

Problems 31-34

### WEEK 9 30m Pre-Algebra - Lesson 42

*Story and Misc. Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.7 p.138

Problems 23-31

### WEEK 10 30m Pre-Algebra - Lesson 43

*Story and Misc. Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.



## Term 2

Sec.7 p.139  
Problems 32-41

### WEEK 10 30m Pre-Algebra - Lesson 44

*Story and Misc. Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.7 p.140

Problems 42-50

### WEEK 11 30m Pre-Algebra - Lesson 45

*Story and Misc. Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.7 p.141

Problems 51-55

### WEEK 11 30m Pre-Algebra - Lesson 46

*Catch-Up Day*

Materials: Hands-On Equations

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 12 30m Pre-Algebra - Lesson 47

*Exam Week*

Materials: Hands-On Equations

#### → EXAMS

Answer question(s) related to course.

Questions will come from:

Hands-On Equations

### WEEK 12 30m Pre-Algebra - Lesson 48

*Exam Week*

Materials: Hands-On Equations

#### → EXAMS

Answer question(s) related to course.

Questions will come from:

Hands-On Equations



## Term 3

### WEEK 1 □ 30m Pre-Algebra - Lesson 49

*Adding and Subtracting Positive and Negative Numbers*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 17 p.2-5

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 17

### WEEK 1 □ 30m Pre-Algebra - Lesson 50

*Simple Algebraic Equations*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 18 p.5-6

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 18

### WEEK 2 □ 30m Pre-Algebra - Lesson 51

*Adding Zero*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 19 p.7-9

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 19

### WEEK 2 □ 30m Pre-Algebra - Lesson 52

*Working with Green and Red Cubes*

□ Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 20 p.9-10

#### → PRACTICE



## Term 3

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 20

### WEEK 3 30m Pre-Algebra - Lesson 53

*Solving More Complex Equations*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 21 p.10-12

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 21

### WEEK 3 30m Pre-Algebra - Lesson 54

*Parentheses*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 22 p.12-14

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 22

### WEEK 4 30m Pre-Algebra - Lesson 55

*Subtracting with Parentheses*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 23 p.14-15

#### → PRACTICE

Complete Classwork Sheet and check solutions.

Classwork Sheets

Lesson 23

### WEEK 4 30m Pre-Algebra - Lesson 56

*Solving Equations with  $-x$*

Materials: Hands-On Equations

#### → DEMONSTRATE

Work through examples and discuss.

Level III

Lesson 24 p.15-16





## Term 3

### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 24

### WEEK 5 30m Pre-Algebra - Lesson 57

*Pictorial Notation*

Materials: Hands-On Equations

### → DEMONSTRATE

Work through examples and discuss.  
Level III  
Lesson 25 p.16-18

### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 25

### WEEK 5 30m Pre-Algebra - Lesson 58

*Traditional Written Notation*

Materials: Hands-On Equations

### → DEMONSTRATE

Work through examples and discuss.  
Level III  
Lesson 26 p.19-22

### → PRACTICE

Complete Classwork Sheet and check solutions.  
Classwork Sheets  
Lesson 26

### WEEK 6 30m Pre-Algebra - Lesson 59

*Number Problems*

Materials: Hands-On Equations Verbal Problems

### → PRACTICE

Solve problems and check solutions.  
Sec.2 p.28  
Problems 25-28

### WEEK 6 30m Pre-Algebra - Lesson 60

*Number Problems (continued)*

Materials: Hands-On Equations Verbal Problems

### → PRACTICE

Solve problems and check solutions.  
Sec.2 p.28  
Problems 29-32



## Term 3

### WEEK 7 30m Pre-Algebra - Lesson 61

*Consecutive Number Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.3 p.44

Problems 20-24

### WEEK 7 30m Pre-Algebra - Lesson 62

*Consecutive Number Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.3 p.44

Problems 25-29

### WEEK 8 30m Pre-Algebra - Lesson 63

*Age Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.4 p.60

Problems 26-31

### WEEK 8 30m Pre-Algebra - Lesson 64

*Age Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.4 p.60

Problems 32-36

### WEEK 9 30m Pre-Algebra - Lesson 65

*Coin Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.

Sec.5 p.80

Problems 32-36

### WEEK 9 30m Pre-Algebra - Lesson 66

*Coin Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.



## Term 3

Sec.5 p.80-81  
Problems 37-41

### WEEK 10 30m Pre-Algebra - Lesson 67

*Coin Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.  
Sec.5 p.81  
Problems 42-45

### WEEK 10 30m Pre-Algebra - Lesson 68

*Distance Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.  
Sec.6 p.106  
Problems 35-39

### WEEK 11 30m Pre-Algebra - Lesson 69

*Distance Problems (continued)*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.  
Sec.6 p.106-107  
Problems 40-44

### WEEK 11 30m Pre-Algebra - Lesson 70

*Story and Misc. Problems*

Materials: Hands-On Equations Verbal Problems

#### → PRACTICE

Solve problems and check solutions.  
Sec.7 p.142  
Problems 56-62

### WEEK 12 30m Pre-Algebra - Lesson 71

*Exam Week*

Materials: Hands-On Equations

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Hands-On Equations



## Term 3

### WEEK 12 30m Pre-Algebra - Lesson 72

*Exam Week*

 Materials: Hands-On Equations

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Hands-On Equations

# Pre-Algebra

## Examination

- Term 1** · Hands-On Equations Verbal Problems Book: p.137 (#16-22)
- Term 2** · Hands-On Equations Verbal Problems Book: p.176-177 (#1-10)
- Term 3** · Hands-On Equations Verbal Problems Book: p.143 (#63-69)